

Homework 4

Due Friday, May 22

The goal of this part of the assignment is to apply the Capital Asset Price Model to cryptocurrencies. In class we used only Bitcoin and Ether. In this assignment, you will also use Cardano and Dogecoin.

1. Download data from yFinance for Bitcoin, Ether, Cardano, Dogecoin, S&P500 and Treasury Bond yields, from January 1, 2020 until May 20, 2026.
2. Construct a portfolio with an initial allocation of 25% Bitcoin, 25% Ether, 25% Cardano, 25% Dogecoin.
3. Compute average returns and standard deviation of returns for (a) Bitcoin, (b), Ether, (c) Cardano, (d) Dogecoin, (e) Portfolio with 25% initial allocation of the four cryptocurrencies, (f) S&P500.
4. Using the Treasury bond yields, which are provided annualized and in percentage, compute the daily mean returns in decimal form (not percentage). Compute the average daily return for the sample period. This is the risk-free rate r_f
5. Compute the betas for (a) Bitcoin, (b) Ether, (c) Cardano, (d) Dogecoin, (e) Portfolio with 25% initial allocation of the four cryptocurrencies. In doing so assume that the expected market return is the average return of the S&P500 over the sample period, and the risk-free rate is the average yield on government bonds (calculated in the previous question).
6. Compare the average returns of (a) Bitcoin, (b) Ether, (c) Cardano, (d) Dogecoin, (e) Portfolio with 25% initial allocation of the four

cryptocurrencies, to the expected returns from the Capital Asset Price model (CAPM).

Please submit the notebook file containing the codes by email: quadrini@usc.edu. Note that you need to download the .ipynb file from Colab.